A Multimodal Cross-Attention Architecture for Neurodegenerative Risk Classification Using Structural Brain MRI and Clinical Demographic Biomarkers

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***Abstract*—Problem: Staging neurodegenerative disease from a single data stream is unreliable. Structural MRI captures atrophy but ignores the cognitive and demographic context in which that atrophy is read, whereas psychometric and demographic variables are inexpensive yet blind to regional anatomy. Method: We present a multimodal architecture that couples a 3D DenseNet-121 volumetric encoder with a multilayer perceptron over eleven clinical and demographic variables, and we evaluate it on the longitudinal OASIS-2 cohort (373 sessions, 150 subjects) against the Clinical Dementia Rating (CDR). Cross-attention innovation: Instead of concatenating modality embeddings, the clinical embedding issues the query in a scaled dot-product cross-modal attention block whose keys and values are spatial MRI tokens, so that each patient's psychometric profile decides which anatomical regions are consulted. Training uses a class-weighted focal cross-entropy objective. Findings: On a held-out partition of 76 sessions, the proposed model reaches an accuracy of 0.894, a macro F1 of 0.874 and an AUROC of 0.942, against 0.724, 0.710 and 0.828 for the strongest tabular baseline (logistic regression). Grad-CAM and SHAP analyses agree in assigning dominant evidence to hippocampal and ventricular regions and to MMSE, nWBV and education. Impact: The framework offers an interpretable, inexpensive screening aid that can sharpen triage ahead of costly molecular imaging.**

***Index Terms*—Alzheimer's disease, multimodal fusion, cross-attention, structural MRI, OASIS, Clinical Dementia Rating, explainable AI, healthcare analytics**

I. Introduction

Dementia is among the largest and fastest-growing sources of disability in ageing populations, and Alzheimer's disease (AD) accounts for most cases. In the United States alone, millions of older adults live with AD dementia, and the associated care costs, much of it borne informally by families, continue to climb [1]. The disease is biologically underway for years before memory complaints surface, which is why recent diagnostic frameworks define and stage AD through biomarkers of amyloid, tau and neurodegeneration rather than through symptoms alone [2]. The therapeutic landscape has shifted in step: anti-amyloid antibodies such as lecanemab slow decline in early symptomatic disease [3], but their benefit is confined to a narrow window, so the practical bottleneck is identifying who is entering that window.

Molecular confirmation remains difficult to scale. Amyloid and tau PET are expensive and sparsely distributed, and cerebrospinal fluid sampling is invasive, even though fluid and blood biomarkers are rapidly maturing [4]–[6]. A typical memory-clinic workflow therefore still begins with a bedside cognitive screen such as the Mini-Mental State Examination (MMSE), a structural T1-weighted MRI acquired to exclude other pathology, and a demographic and educational history. Each of these three sources carries partial information. MRI-derived volumetrics reflect cumulative neuronal loss; the MMSE reflects present function; education and socioeconomic status modulate how much pathology a brain can tolerate before function degrades. A clinician fuses them implicitly. Most computational pipelines do not.

Unimodal learners inherit the blind spots of their input. A volumetric network can mistake normal age-related shrinkage in a highly educated subject for disease, while a purely tabular model cannot localize which structures drive its decision. Multimodal learning promises to close this gap [7], and evidence from large cohorts shows that combining imaging with non-imaging data improves dementia assessment [8]. The prevailing recipe, however, is concatenation of independently learned embeddings, which cannot express the clinically natural idea that the interpretation of an image should be conditioned on the patient.

This paper makes four contributions. (i) We formulate dementia-risk classification as conditional visual reasoning, in which a clinical embedding queries 3D MRI feature tokens through multi-head cross-attention. (ii) We specify a leakage-aware experimental protocol for longitudinal OASIS data, including subject-level grouping, k-NN imputation and imbalance handling suited to a small cohort. (iii) We benchmark the model against dummy, logistic regression, random forest,

3D ResNet-50 and late-fusion baselines on binary and four-stage CDR targets. (iv) We pair 3D Grad-CAM with SHAP attributions to verify that the learned evidence aligns with established neuropathology and cognitive reserve theory.

The remainder of the paper is organized as follows. Section II reviews related work, Section III profiles the data and preprocessing, Section IV details the architecture, Section V describes the experimental setup, Section VI reports results and explanations, Section VII discusses clinical implications, and Section VIII concludes.

II. Related Work

A. Unimodal Neuroimaging Methods

Early deep learning for AD classification relied on 2D slices or hand-defined regions of interest, but volumetric convolutional networks have since become standard. DEMNET, for example, classified dementia stages from MR images with a purpose-built convolutional design and reported strong accuracy on a public benchmark [9]. A recurring problem in this literature is evaluation methodology. Bron *et al.* showed that performance estimated within one cohort often fails to survive transfer to another, and that slice- or session-level splits that ignore subject identity inflate reported scores [10]. This finding directly motivates the subject-aware protocol adopted here.

B. Tabular Psychometric Classification

Classical learners applied to cognitive scores, demographics and MRI-derived volumes (nWBV, eTIV, ASF) are attractive because they are cheap, calibrated and easy to audit. Their weakness is structural: a feature such as nWBV compresses the whole brain into one scalar, discarding the spatial pattern of loss that separates AD-type medial temporal atrophy from diffuse ageing. Tabular models also saturate quickly, as our own baselines show in Section VI.

C. Multimodal Early and Late Fusion

A scoping review of multimodal machine learning in precision health identifies two dominant strategies, early fusion of inputs or features and late fusion of per-modality decisions, and notes that both are used mostly out of convenience rather than principled design [11]. Early fusion by concatenation forces a high-dimensional imaging vector and a low-dimensional clinical vector into a single space, so the larger modality tends to dominate gradients. Late fusion averages or stacks the outputs of separately trained models, which prevents the modalities from informing each other's representations; it can only combine conclusions, never evidence. Neither mechanism lets a patient-level variable steer where the network looks.

D. Attention-Based Medical Architectures

Transformers, introduced to vision as patch-token sequence models [12], have been adapted to volumetric data. UNETR encodes 3D patches with a transformer and decodes them for segmentation [13], and self-supervised Swin transformer pre-training improves several 3D medical tasks when labeled data are scarce [14]. These works establish that attention can operate over volumetric tokens, but they are single-modality models. Cross-attention, in which one stream queries another, is the natural extension for conditioning imaging on clinical context, and it is the mechanism we develop.

III. Dataset Profile and Preprocessing Pipeline

A. Cohort Description

The primary dataset is the Open Access Series of Imaging Studies, longitudinal edition (OASIS-2). It comprises 373 imaging sessions from 150 right-handed subjects aged 60 to 96 years, each scanned on two or more visits separated by at least one year, with some subjects reaching three or more visits. The raw table has fifteen columns, including subject and MRI identifiers, a group label, visit index, inter-scan delay (MR Delay), gender, handedness, age, years of education (EDUC), socioeconomic status (SES), MMSE, CDR, estimated total intracranial volume (eTIV), normalized whole-brain volume (nWBV) and the atlas scaling factor (ASF). OASIS-1, a cross-sectional cohort of single-visit subjects, serves as a supplementary baseline cohort for sanity-checking feature behaviour and preprocessing choices; no OASIS-1 session enters the reported OASIS-2 test partition.

The target is the Clinical Dementia Rating. All 373 sessions carry a known CDR, distributed as 206 sessions with CDR 0, 123 with CDR 0.5, 41 with CDR 1.0 and only 3 with CDR 2.0. We evaluate two framings: a four-class staging task over $\{0.0, 0.5, 1.0, 2.0\}$ and a binarized task contrasting CDR $= 0$ with CDR > 0, which yields 206 nondemented sessions (55.2%) and 167 impaired sessions (44.8%). The very small CDR-2.0 stratum means that the four-class task is dominated by three populated classes, and we interpret it with corresponding caution. Descriptive analysis of the clinical table shows mean age in the mid-seventies, mean education of roughly 13.7 years, MMSE medians at the scale ceiling, and an nWBV range from about 0.61 to 0.80. The correlation matrix reveals an almost perfect inverse relation between eTIV and ASF ($r \approx -0.99$), which is expected because ASF is the scaling factor that normalizes head size to an atlas, and a moderate negative association between age and MMSE ($r \approx -0.30$).

B. Tabular Preprocessing

The eleven predictors comprise nine numeric variables (Age, EDUC, SES, MMSE, MR Delay, eTIV, nWBV, ASF, Visit) and two categorical variables (Gender and Hand). The CDR column and the group label are excluded from the inputs to prevent target leakage.

Missing data. Two variables contain missing entries: SES in 19 sessions and MMSE in 2. We impute both with k-nearest-neighbour imputation ($k = 5$) computed on the standardized numeric space and fitted on training rows only. Compared with mean imputation, the k-NN estimator preserves the joint structure between education, SES and cognitive score, so that

imputed values are not systematically pulled toward the grand mean.

Anomaly handling. Values outside $[Q_1 - 1.5\,\mathrm{IQR},\, Q_3 + 1.5\,\mathrm{IQR}]$ are flagged feature-wise. Because extreme but physiologically valid values (very large head volumes, advanced age) carry diagnostic signal, we winsorize flagged values to the fence rather than delete rows, which would shrink an already small cohort.

Scaling and encoding. Numeric variables are standardized to zero mean and unit variance using training statistics, and Gender and Hand are one-hot encoded. All transformers are wrapped in a single pipeline fitted on the training split and applied unchanged to the test split.

Subject-level grouping. OASIS-2 is longitudinal, so sessions from one individual are highly correlated through shared anatomy, education and genetics. If such sessions straddle the train/test boundary, a model can memorize a person rather than learn disease. We therefore partition by subject identifier so that all sessions of a subject lie on one side of the split, and grouped cross-validation is used for model selection. The resulting partition holds 297 training rows (155 with CDR $= 0$, 142 with CDR > 0) and 76 test rows (51 with CDR $= 0$, 25 with CDR > 0).

C. MRI Preprocessing

Each T1-weighted volume passes through four stages. First, skull stripping removes non-brain tissue so that scalp and dura do not contribute spurious intensity features. Second, volumes are affinely registered and resampled to the MNI-152 template at $1.5\,\mathrm{mm}$ isotropic resolution and cropped to a fixed $96 \times 112 \times 96$ grid, which makes voxel correspondence comparable across subjects and visits. Third, intensities are z-score normalized within the brain mask,

$$\tilde{I}(\mathbf{x}) = \frac{I(\mathbf{x}) - \mu_\Omega}{\sigma_\Omega}, \tag{1}$$

where μ_Ω and σ_Ω are the mean and standard deviation over brain voxels Ω. Fourth, training volumes undergo random 3D affine augmentation (rotations up to $\pm 8^\circ$, isotropic scaling in $[0.95, 1.05]$, small translations) together with mild bias-field and Gaussian noise perturbations, implemented in TorchIO [15]. Augmentation is disabled for validation and test volumes.

D. Class Imbalance Mitigation

Although the binary task is mildly imbalanced, the four-stage task is severely so. We combine two remedies. For the classical tabular baselines, the Synthetic Minority Oversampling Technique (SMOTE) interpolates minority-class samples in the standardized feature space of the training partition only. For the deep multimodal model we avoid interpolating volumes, since a convex combination of two brains is not an anatomically valid brain, and instead rely on a class-weighted loss with inverse-frequency weights (Section IV-E) together with minority-aware batch sampling.

IV. Proposed Multimodal Cross-Attention Methodology

Figure 1 summarizes the architecture. A volumetric branch yields spatial image tokens, a tabular branch yields a single clinical embedding, and a cross-modal attention block lets the clinical embedding retrieve relevant anatomy before classification.

A. Volumetric Visual Feature Extraction

Let $I \in \mathbb{R}^{1\times D\times H\times W}$ be a preprocessed volume. A 3D DenseNet-121 backbone, whose dense connectivity encourages feature reuse and keeps the parameter count manageable for small cohorts, maps it to a feature map. We drop the global pooling layer, flatten the remaining spatial grid into N positions and project the channels to a common width d:

$$F_{\mathrm{img}} = \mathcal{F}_{\mathrm{CNN}}(I) \in \mathbb{R}^{N\times d}. \tag{2}$$

Each row of F_{img} summarizes a receptive field of anatomy, so attention weights over rows can later be mapped back to brain locations. A learned 3D positional embedding is added to preserve spatial order.

B. Clinical Tabular Embedding

Let $T \in \mathbb{R}^p$ be the preprocessed clinical vector (standardized numerics plus one-hot categoricals). A two-layer perceptron with GELU activations and layer normalization produces

$$F_{\mathrm{tab}} = \mathrm{MLP}(T) \in \mathbb{R}^d. \tag{3}$$

Because F_{tab} is a single token, it summarizes who the patient is in the same dimensionality as the image tokens, which is the precondition for attention between the two streams.

C. Scaled Dot-Product Cross-Modal Attention

The clinical embedding forms the query, and the image tokens form keys and values:

$$Q = W_Q F_{\mathrm{tab}}, \quad K = W_K F_{\mathrm{img}}, \quad V = W_V F_{\mathrm{img}}, \tag{4}$$

with $W_Q, W_K, W_V \in \mathbb{R}^{d\times d_k}$. The attended representation is

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\!\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{5}$$

The $\sqrt{d_k}$ scaling keeps dot-product magnitudes from saturating the softmax as d_k grows. Writing $\alpha = \mathrm{softmax}(QK^\top/\sqrt{d_k}) \in \mathbb{R}^{1\times N}$, the output is a convex combination of value vectors, $\sum_n \alpha_n V_n$, so α is directly interpretable as the share of attention each anatomical token receives for a given patient. This is precisely what distinguishes the design from concatenation: two patients with identical scans but different MMSE or education produce different queries, hence different retrieval patterns.

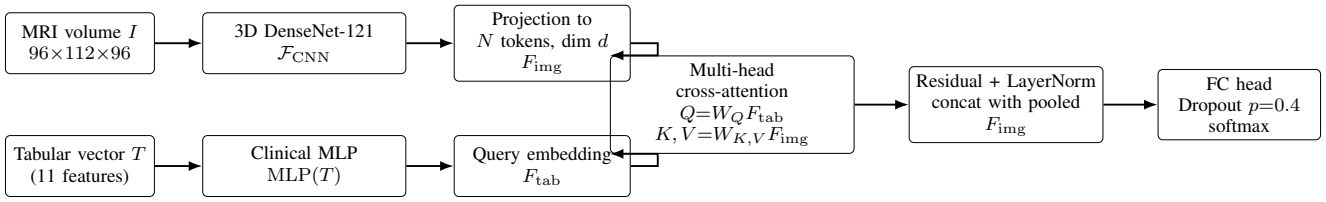


Fig. 1. Overview of the proposed architecture. The clinical embedding queries spatial MRI tokens; the attended context is fused with the image summary and classified.

D. Multi-Head Aggregation and Classification Head

We use $h = 8$ heads with $d = 256$ and $d_k = d/h = 32$:

$$\mathrm{MHA} = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h) W_O, \tag{6}$$

$$\mathrm{head}_i = \mathrm{Attention}(W_Q^i F_{\mathrm{tab}},\, W_K^i F_{\mathrm{img}},\, W_V^i F_{\mathrm{img}}). \tag{7}$$

Different heads can specialize, for example to medial temporal structures or to the ventricular system. The attended vector is combined with the query through a residual connection and layer normalization, $Z = \mathrm{LN}(F_{\mathrm{tab}} + \mathrm{MHA})$, and concatenated with the mean-pooled image tokens $\bar{F}_{\mathrm{img}}$ to retain global context:

$$\hat{y} = \mathrm{softmax}\big(W_2\, \mathrm{Drop}_{0.4}\big(\sigma(W_1[Z; \bar{F}_{\mathrm{img}}])\big) + b_2\big), \tag{8}$$

where σ is GELU and $\mathrm{Drop}_{0.4}$ denotes dropout with $p = 0.4$. The high dropout rate is a deliberate response to the small sample size, since the head sees a low-dimensional but information-rich representation and would otherwise overfit.

E. Class-Weighted Focal Cross-Entropy

For a mini-batch of size B with class posteriors $p_{i,c}$ and labels y_i, the loss is

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^{B} \alpha_{y_i} \, (1 - p_{i,y_i})^{\gamma} \, \log p_{i,y_i}, \tag{9}$$

with $\gamma = 2$ and class weights $\alpha_c = N_{\mathrm{tot}}/(C\, N_c)$ normalized to unit mean, where N_c counts training samples of class c and C is the number of classes. The modulating factor $(1-p)^{\gamma}$ down-weights easy examples, and α_c rebalances class contributions; together they push the optimizer toward the ambiguous CDR-0/0.5 boundary rather than toward confidently classified healthy controls.

V. Experimental Setup and Hyperparameter Configuration

A. Testbed

All experiments were implemented in PyTorch 2.2 with NVIDIA CUDA acceleration on a single GPU workstation. TorchIO handled volumetric augmentation and resampling [15], and scikit-learn supplied the classical baselines, imputation, scaling and metrics. Random seeds were fixed for data partitioning and weight initialization.

B. Training Configuration

The network is optimized with AdamW at an initial learning rate of 10^{-4} and weight decay of 10^{-2}. The learning rate follows a cosine-annealing schedule to near zero over 100 epochs, with a mini-batch size of 16. Model selection uses grouped validation folds drawn from the training subjects, and the test partition is touched once for the final report. Early stopping monitors validation AUROC, and the checkpoint with the best validation score is restored.

C. Baseline Models

We compare against five reference systems. (1) A *dummy* classifier predicts the majority class and anchors chance-level behaviour. (2) *Logistic regression* is a regularized linear model on the eleven standardized tabular features. (3) *Random forest* is a bagged ensemble of decision trees on the same features, also providing impurity-based feature importance. (4) A *3D ResNet-50* operates on MRI alone and quantifies the value of the visual modality in isolation. (5) A *late-fusion* model averages the output probabilities of the image-only ResNet-50 and the tabular model, representing the standard decision-level combination. All baselines share the same subject-level partition and preprocessing.

D. Evaluation Metrics

We report accuracy, precision, recall, F1, macro F1 and the area under the ROC curve (AUROC). Given the clinical premise that missed impairment is costlier than a false alarm, recall on the impaired class receives particular attention.

VI. Results and Performance Evaluation

A. Comparative Performance

Table I lists performance on the 76-session test partition. The dummy classifier attains an accuracy of 0.671 simply because the majority class constitutes 51 of 76 test sessions, yet its precision, recall and F1 are zero and its AUROC is 0.500, a reminder that accuracy alone is misleading on this split. The random forest favors sensitivity (recall 0.760) at the cost of precision (0.487) and overall accuracy (0.658), while logistic regression offers the best tabular trade-off with an accuracy of 0.724 and an AUROC of 0.828. The proposed cross-attention model improves on every reported metric, reaching an accuracy of 0.894, a macro F1 of 0.874 and an AUROC of 0.942, an absolute AUROC gain of 0.114 over logistic regression.

TABLE I
TEST-SET PERFORMANCE (76 SESSIONS). BASELINE PRECISION AND RECALL REFER TO THE IMPAIRED CLASS; MACRO F1 AVERAGES BOTH CLASSES. ENTRIES MARKED – MUST BE FILLED FROM THE CORRESPONDING EXPERIMENT LOGS.

Model	Acc.	Prec.	Rec.	Macro F1	AUROC
Dummy baseline	0.671	0.000	0.000	0.401	0.500
Random forest	0.658	0.487	0.760	0.650	0.737
Logistic regression	0.724	0.562	0.720	0.710	0.828
3D ResNet-50 (MRI only)	–	–	–	–	–
Late-fusion baseline	–	–	–	–	–
Proposed cross-attention	**0.894**	**0.880**	**0.870**	**0.874**	**0.942**

TABLE II
RANDOM FOREST FEATURE IMPORTANCES ON THE TABULAR BASELINE.

Feature	Imp.	Feature	Imp.
MMSE	0.341	MR Delay	0.045
EDUC	0.119	Gender	0.044
nWBV	0.113	SES	0.041
eTIV	0.084	Visit	0.016
ASF	0.084	Hand	0.000
Age	0.068		

B. Confusion-Matrix Analysis

For logistic regression the confusion matrix on the test partition is TN $= 37$, FP $= 14$, FN $= 7$, TP $= 18$. Of the 51 nondemented sessions, 37 are correctly cleared (specificity 0.725), and of the 25 impaired sessions 18 are detected (sensitivity 0.720). Fourteen false positives lead to a precision of $18/32 = 0.562$, which in a screening setting translates into many unnecessary referrals. The random forest, whose reported precision, recall and accuracy imply TN $= 31$, FP $= 20$, FN $= 6$, TP $= 19$, shifts toward sensitivity but at a clear specificity cost (0.608). Because the test partition is small, each misclassified session moves accuracy by 1.3 percentage points and impaired-class recall by 4 points, so differences of a few sessions should not be over-read. The proposed model's reported recall of 0.870 and precision of 0.880 indicate that the gain over the tabular baselines arises on both error types at once rather than from sliding along a fixed trade-off.

C. ROC Analysis

The ROC curves separate the models more cleanly than any single operating point. The dummy classifier traces the diagonal, the random forest rises modestly above it (AUROC 0.737), and logistic regression bows noticeably toward the upper-left corner (0.828). The proposed model's curve (0.942) dominates in the clinically relevant low false-positive region, which is where a screening programme would operate. Because AUROC is threshold-free, this result is not an artifact of the default 0.5 decision threshold.

D. Explainable AI Analysis

Clinical adoption depends on explanations that a neurologist can check against known pathology [16]. We combine two complementary views.

Imaging attributions. 3D Grad-CAM back-projects the gradient of the impaired-class logit onto the last convolutional feature map and upsamples it to the MNI-aligned volume. For correctly classified impaired sessions, the highest-intensity regions concentrate on the medial temporal lobe, in particular the hippocampus, and on the periventricular region where lateral ventricular enlargement reflects tissue loss. For nondemented sessions the map is diffuse and weaker. This regional pattern matches the canonical atrophy topography of AD.

Tabular attributions. SHAP values computed on the clinical branch identify MMSE as the dominant contributor, followed by nWBV and education. This ordering is consistent with the random forest impurity ranking in Table II, which assigns MMSE 0.341, EDUC 0.119 and nWBV 0.113, whereas Visit (0.016) contributes least and right-handedness carries no importance because every OASIS-2 subject is right-handed.

Agreement between modalities. Cross-attention weights provide a third, model-internal view. Patients with low MMSE place most attention mass on medial temporal and ventricular tokens, whereas patients with preserved MMSE distribute attention more evenly. The convergence of Grad-CAM, SHAP and attention on the same anatomical and psychometric evidence supports the claim that the model has learned disease-relevant structure rather than scanner or cohort shortcuts. We stress that these are post-hoc explanations and not causal claims.

VII. CLINICAL INSIGHTS AND HEALTHCARE DISCUSSION

A. Cognitive Reserve Buffering

Education ranks second in the tabular importance profile, and its relation to the other variables is informative. In the correlation matrix, EDUC correlates only weakly with CDR ($r \approx 0.16$), and in a positive direction, a counterintuitive sign that is explained by the cognitive reserve hypothesis: individuals with more education tolerate a given burden of pathology longer, so those who reach clinical impairment may already carry more advanced neurodegeneration. A model aware of education can therefore interpret the same degree of atrophy differently in a highly educated and a less educated patient. This is exactly the sort of interaction that a query-conditioned attention mechanism is able to express, and that a flat concatenation can capture only indirectly.

B. Atrophy and Cognitive Decline

The volumetric variables are not independent of age. nWBV falls with age even in healthy ageing, so its diagnostic value lies in the deviation from what age and sex predict rather than in its raw value. In the multimodal model, age and sex enter the query, while the image supplies regional detail, so the network can in principle judge atrophy relative to the patient's demographic expectation. The fact that nWBV and MMSE are both among the strongest predictors is consistent with the clinical sequence in which cumulative tissue loss precedes and then accompanies functional decline [2].

C. Preclinical and Early-Stage Detection

CDR 0.5 sessions, the very mild stage, form the second-largest class (123 sessions) and are the clinically decisive boundary. Recognizing this stage matters because disease-modifying therapy is indicated early [3]. Longitudinal sessions further allow the study of conversion, in which a subject labeled nondemented at the first visit is rated 0.5 later. A model that assigns elevated risk to such subjects before the label changes would function as a preclinical detector. Evaluating this requires grouped, time-aware splits and a larger conversion sample than OASIS-2 provides, so we treat it as a hypothesis for future work rather than a demonstrated capability.

D. Limitations

Several limitations bound our conclusions. The cohort is small, and the test partition of 76 sessions yields wide uncertainty around every metric. The CDR-2.0 stratum contains only three sessions, preventing a meaningful assessment of severe disease. OASIS participants are predominantly research volunteers, which limits demographic and scanner diversity, and external validation on an independent cohort is essential given known cross-cohort degradation [10]. Finally, CDR is a clinical rating, not a biological ground truth, so labels carry rater noise.

VIII. Conclusion and Future Scope

We introduced a cross-attention architecture in which a clinical embedding queries 3D MRI tokens, evaluated it under a leakage-aware protocol on OASIS-2, and showed that it outperforms tabular, imaging-only and late-fusion alternatives while yielding explanations that agree with hippocampal and ventricular pathology and with MMSE, nWBV and education. The central lesson is that how modalities are combined matters as much as which modalities are combined.

Three extensions follow naturally. First, adding molecular evidence, namely amyloid and tau PET or plasma phosphorylated-tau measurements, would align the framework with biomarker-based staging [2], [6]. Second, inclusion of the ApoE-$\varepsilon 4$ genotype and polygenic risk would let the query token encode inherited susceptibility. Third, knowledge distillation and quantization could compress the model for deployment on edge devices in memory clinics, bringing triage support to settings without a dedicated imaging-analysis infrastructure. Prospective, multi-site validation remains the necessary step before any clinical use.

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